Problem 3.

Consider the polynomial

(1) $$p(x,y) = 2 + 5x - 3y + x^2 - 7xy.$$

Show that $\forall$ point (x_0, y_0), The Taylor series for $p(x,y)$ (Let us call it $P(x,y)$, about (x_0, y_0) is $p(x,y)$).

Sol-n:

So, based on the prob-m statement,

(2) $$P(x,y) = \sum_{|\alpha| \geq 0} \frac{D^\alpha p(\underline{x}_0)}{\alpha!} (\underline{x} - \underline{x}_0)^\alpha,$$ the

Taylor series of $p(x,y)$ about $\underline{x} = \underline{x}_0$ pt.

Want to show (WTS): $P(x,y) \equiv p(x,y)$.

Let $\underline{x} = \langle x, y \rangle \overset{\text{or}}{=} \begin{bmatrix} x \\ y \end{bmatrix}$ & $\underline{x}^0 = \langle x_0, y_0 \rangle = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$, an arbitrary position vector.

Using the notations in the previous Pr. 2, let me calculate α's first. Note that the given polyn-L $p(x,y)$ in (1) is of 2nd degree, so <u>it is enough to consider</u> the α's s.t

(3) $$0 \leq |\alpha| \leq \overset{=(2+1)}{3}$$

because when trying to find the successive derivatives of 2nd degree polynomial, the derivatives of order ≥ 3 becomes zero. i.e. Consider

(4) $$f(x) = x^2 \Rightarrow f'(x) = 2x \Rightarrow f''(x) = 2 \Rightarrow \overset{'''}{f}(x) = 0.$$

(PS: indeed $0 \leq |\alpha| \leq 2$ should work too, but to really <u>show</u> the rest becomes zero, we got $0 \leq |\alpha| \leq 3$.).

Thus, α's have the form:

(5) $$\alpha = \overset{\alpha_1\,\alpha_2}{(0,0)},\ \overset{\alpha_1\,\alpha_2}{(1,0)},\ \overset{\alpha_1\,\alpha_2}{(0,1)},\ \overset{\cdots}{(1,1)},\ (2,0),\ (0,2),\ (2,1),\ (1,2),\ (0,3),\ (3,0).$$

Then, by (2), Taylor series of 3rd order:

(6) $$P(x,y) = \sum_{0\le|\alpha|\le 3} \frac{D^{\alpha}p(\underline{x}_0)}{\alpha!}(\underline{x}-\underline{x}_0)^{\alpha} =$$

$$= \frac{1}{(0,0)!}D^{(0,0)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(0,0)} + \frac{1}{(1,0)!}D^{(1,0)}p(\underline{x}_0)\cdot(\underline{x}-x_0)^{(1,0)} +$$

$$+ \frac{1}{(0,1)!}D^{(0,1)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(0,1)} + \frac{1}{(1,1)!}D^{(1,1)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(1,1)} +$$

$$+ \frac{1}{(2,0)!}D^{(2,0)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(2,0)} + \frac{1}{(0,2)!}D^{(0,2)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(0,2)} +$$

$$+ \frac{1}{(2,1)!}D^{(2,1)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(2,1)} + \frac{1}{(1,2)!}D^{(1,2)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(1,2)} +$$

$$+ \frac{1}{(0,3)!}D^{(0,3)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(0,3)} + \frac{1}{(3,0)!}D^{(3,0)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(3,0)}.$$

Now let me compute each term separately for

(1) $$p(x,y) = 2+5x-3y+x^2-7xy$$

(7) $$\frac{1}{(0,0)!}\cdot D^{(0,0)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(0,0)} = \frac{1}{0!\,0!}\cdot\frac{\partial^{0+0}p(x_0,y_0)}{\partial x^0\,\partial y^0}\cdot(x-x_0)^0\cdot(y-y_0)^0 =$$

$$= (1)\cdot p(x_0,y_0)\cdot(1)\cdot(1) = 2+5x_0-3y_0+x_0^2-7x_0\cdot y_0.$$

(8) $$\frac{1}{(1,0)!}D^{(1,0)}p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(1,0)} = (1)\cdot\frac{\partial^{1+0}p(x_0,y_0)}{\partial x^1\,\partial y^0}\cdot(x-x_0)^1\,(y-y_0)^0 =$$

$$= (1)\cdot\frac{\partial p(x_0,y_0)}{\partial x}\cdot(x-x_0)\cdot(1) = \frac{\partial}{\partial x}\left[2+5x-3y+x^2-7xy\right]_{(x_0,y_0)}\cdot(x-x_0) =$$

$$= \left[5+2x-7y\right]_{(x_0,y_0)} \cdot (x-x_0) = (5+2x_0-7y_0)(x-x_0) \quad .$$

(9) $$\frac{1}{(0,1)!} D^{(0,1)} p(\underline{x}_0)(\underline{x}-\underline{x}_0)^{(0,1)} = (1)\cdot \frac{\partial^{0+1} p(x_0,y_0)}{\partial x^0 \partial y^1} \cdot (x-x_0)^0 \cdot (y-y_0)^1 =$$

$$= \frac{\partial p(x_0,y_0)}{\partial y} \cdot (1) \cdot (y-y_0) = \frac{\partial}{\partial y}\left[2+5x-3y+x^2-7xy\right]_{(x_0,y_0)} \cdot (y-y_0) =$$

$$= \left[-3-7x\right]_{(x_0,y_0)} \cdot (y-y_0) = (-3-7x_0)\cdot(y-y_0).$$

(10) $$\frac{1}{(1,1)!} D^{(1,1)} p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(1,1)} = (1)\cdot \frac{\partial^{1+1} p(x_0,y_0)}{\partial x^1 \partial y^1} \cdot (x-x_0)^1 \cdot (y-y_0)^1 =$$

$$= \frac{\partial^2 p(x_0,y_0)}{\partial x \partial y} \cdot (x-x_0)\cdot(y-y_0) = \frac{\partial}{\partial x}\overbrace{\left[\frac{\partial p(x,y)}{\partial y}\right]}^{\text{by (9)}}_{(x_0,y_0)} \cdot (x-x_0)\cdot(y-y_0) =$$

$$= \frac{\partial}{\partial x}\left[-3-7x\right]_{(x_0,y_0)} \cdot (x-x_0)\cdot(y-y_0) = \left[-7\right]_{(x_0,y_0)} \cdot (x-x_0)\cdot(y-y_0) =$$

$$= (-7)\cdot(x-x_0)\cdot(y-y_0).$$

(11) $$\frac{1}{(2,0)!} D^{(2,0)} p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(2,0)} = \frac{1}{2!\cdot 0!}\cdot \frac{\partial^{2+0} p(x_0,y_0)}{\partial x^2 \cdot \partial y^0} \cdot (x-x_0)^2 \cdot (y-y_0)^0 =$$

$$= \frac{1}{2}\cdot \frac{\partial^2 p(x_0,y_0)}{\partial x^2} \cdot (x-x_0)^2 \cdot (1) = \frac{1}{2}\cdot \frac{\partial}{\partial x}\overbrace{\left[\frac{\partial p(x,y)}{\partial x}\right]}^{\text{by (8)}}_{(x_0,y_0)} \cdot (x-x_0)^2 =$$

$$= \frac{1}{2}\cdot \overbrace{\frac{\partial}{\partial x}\left[5+2x-7y\right]}^{2}_{(x_0,y_0)} \cdot (x-x_0)^2 = (x-x_0)^2.$$

(12) $$\frac{1}{(0,2)!} D^{(0,2)} p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(0,2)} = \frac{1}{0!\cdot 2!}\cdot \frac{\partial^{0+2} p(x_0,y_0)}{\partial x^0 \cdot \partial y^2} \cdot (x-x_0)^0 \cdot (y-y_0)^2 =$$

$$= \frac{1}{2}\cdot \frac{\partial^2 p(x_0,y_0)}{\partial y^2} \cdot (1)\cdot(y-y_0)^2 = \frac{1}{2}\cdot \frac{\partial}{\partial y}\overbrace{\left[\frac{\partial p}{\partial y}\right]}^{\text{by (6)}}_{(x_0,y_0)} \cdot (y-y_0)^2 = \frac{1}{2}\overbrace{\frac{\partial}{\partial y}\left[-3-7x\right]}^{0}_{(x_0,y_0)} \cdot (y-y_0)^2 =$$

$$= 0.$$

(13) $$\frac{1}{(2,1)!} D^{(2,1)} p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(2,1)} = \frac{1}{2!\cdot 1!}\cdot\frac{\partial^{2+1} p(x_0,y_0)}{\partial x^2\cdot\partial y^1}\cdot(x-x_0)^2\cdot(y-y_0)^1 =$$

$$= \frac{1}{2}\cdot\frac{\partial^3 p(x_0,y_0)}{\partial x^2\partial y}\cdot(x-x_0)^2\cdot(y-y_0) = \frac{1}{2}\cdot\frac{\partial}{\partial x}\underbrace{\left[\frac{\partial^2 p(x_0,y_0)}{\partial x\,\partial y}\right]}_{(-7)\text{ by }(10).}\cdot(x-x_0)^2\cdot(y-y_0) =$$

$$= \frac{1}{2}\frac{\partial}{\partial x}\left[-7\right]_{(x_0,y_0)}\cdot(x-x_0)^2\cdot(y-y_0) = \frac{1}{2}\cdot 0\Big|_{(x_0,y_0)}\cdot(x-x_0)^2\cdot(y-y_0)^2 = 0.$$

(14) $$\frac{1}{(1,2)!} D^{(1,2)} p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(1,2)} = \frac{1}{1!\cdot 2!}\cdot\frac{\partial^{1+2} p(x_0,y_0)}{\partial x^1\,\partial y^2}\cdot(x-x_0)^1\cdot(y-y_0)^2 =$$

$$= \frac{1}{2}\cdot\frac{\partial}{\partial x}\underbrace{\left[\frac{\partial p^2(x,y)}{\partial y^2}\right]}_{=0\text{ by }(12)}_{(x_0,y_0)}\cdot(x-x_0)(y-y_0)^2 =$$

$$= \frac{1}{2}\cdot\frac{\partial}{\partial x}\left[0\right]_{(x_0,y_0)}\cdot(x-x_0)\cdot(y-y_0)^2 = 0.$$ (we can see last calc-ns ((12),(13),(14) = 0 as they're 3rd order)

(15) $$\frac{1}{(0,3)!} D^{(0,3)} p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(0,3)} = \frac{1}{0!\cdot 3!}\frac{\partial^{0+3} p(x_0,y_0)}{\partial x^0\partial y^3}(x-x_0)^0\cdot(y-y_0)^3 =$$

$$= \frac{1}{6}\cdot\frac{\partial^3 p(x_0,y_0)}{\partial y^3}\cdot(1)\cdot(y-y_0)^3 = \frac{1}{6}\cdot\frac{\partial}{\partial y}\underbrace{\left[\frac{\partial^2 p}{\partial y^2}\right]}_{=0\text{ by }(12)}_{(x_0,y_0)}\cdot(y-y_0)^3 =$$

$$= \frac{1}{6}\frac{\partial}{\partial y}\left[0\right]_{(x_0,y_0)}\cdot(y-y_0)^3 = 0.$$

(16) $$\frac{1}{(3,0)!} D^{(3,0)} p(\underline{x}_0)\cdot(\underline{x}-\underline{x}_0)^{(3,0)} = \frac{1}{3!\cdot 0!}\frac{\partial^{3+0} p(x_0,y_0)}{\partial x^3\,\partial y^0}\cdot(x-x_0)^3\cdot(y-y_0)^0 =$$

$$= \frac{1}{6}\frac{\partial^3 p(x_0,y_0)}{\partial x^3}\cdot(x-x_0)^3\cdot(1) = \frac{1}{6}\cdot\frac{\partial}{\partial x}\underbrace{\left[\frac{\partial^2 p}{\partial x^2}\right]}_{=2\text{ by }(11)}_{(x_0,y_0)}\cdot(x-x_0)^3 =$$

$$= \frac{1}{6}\cdot\frac{\partial}{\partial x}\left[2\right]_{(x_0,y_0)}\cdot(x-x_0)^3 = \frac{1}{6}\cdot\left[0\right]_{(x_0,y_0)}\cdot(x-x_0)^3 = 0.$$

Of course, then, all $D^\alpha f(\underline{x}_0) \equiv 0$ for all $|\alpha| \geq 3$.

So, substituting the entities in (7) – (16) into (6) gives

(17) $$P(x,y) = \sum_{0 \le |\alpha| \le 3} \frac{D^\alpha p(\underline{x}_0)}{\alpha!} (\underline{x} - \underline{x}_0)^\alpha =$$

$$= (2 + 5x_0 - 3y_0 + x_0^2 - 7x_0 \cdot y_0) + (5 + 2x_0 - 7y_0) \cdot (x - x_0) +$$

$$+ (-3 - 7x_0) \cdot (y - y_0) + (-7) \cdot (x - x_0) \cdot (y - y_0) + (x - x_0)^2 +$$

$$+ 0 + 0 + 0 + 0 + 0 =$$

$$= (2 + 5x_0 - 3y_0 + x_0^2 - 7x_0 y_0) + 5x + \cancel{2x_0 x} - 7y_0 x -$$

$$- 5x_0 - 2x_0^2 + 7x_0 y_0 - 3y - 7x_0 y + 3y_0 + 7x_0 y_0 +$$

$$+ (-7x + 7x_0) \cdot (y - y_0) + x^2 - \cancel{2x_0 x} + x_0^2 =$$

$$= (2 + \cancel{5x_0} - \cancel{3y_0} + x_0^2 - \cancel{7x_0 y_0}) + 5x \boxed{- 7y_0 x} - \cancel{5x_0} - x_0^2$$

$$+ \cancel{14 x_0 y_0} - 3y - 7x_0 y + \cancel{3y_0} - 7xy + 7x_0 y \boxed{+ 7y_0 x} - \cancel{7x_0 y_0}$$

$$+ x^2 = 2 + 5x - 3y - 7xy + x^2$$

So

(18) $$P(x,y) = \sum_{|\alpha| \ge 0} \frac{D^\alpha p(\underline{x}_0)}{\alpha!} (\underline{x} - \underline{x}_0)^\alpha = 2 + 5x - 3y - 7xy + x^2 \equiv$$

$$\equiv 2 + 5x - 3y + x^2 - 7xy = p(x,y).$$

✓

(18) is showing the desired result. ▨